ONLINE EVENT TICKET BOOKING SYSTEM

AI23431- WEBTECHNOLOGY AND MOBILE APPLICATION

PROJECT REPORT

SUBMITTED BY

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AND MACHINE LEARNING

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(AUTONOMOUS),

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RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled "**ONLINE EVENT TICKET BOOKING SYSTEM**" is the Bonafide work of SOWMYA R (2116241501207) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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SOWMYA R - 241501207

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES (PEO'S)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Machine Learning Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVE

- To identify and formulate real-world problems that can be solved using Artificial Intelligence and Machine Learning techniques.
- To apply theoretical and practical knowledge of AI/ML for designing innovative, data-driven solutions.
- To integrate various tools, frameworks, and algorithms to develop, test, and validate AI/ML models.
- To demonstrate effective teamwork, project management, and communication skills through collaborative project execution.
- To instill awareness of ethical, societal, and environmental considerations in the design and deployment of intelligent systems.

COURSE OUTCOME

- **CO 1:** Analyze and define a real-world problem by identifying key challenges, project requirements and constraints.
- **CO 2:** Conduct a thorough literature review to evaluate existing solutions, identify research gaps and formulate research questions.
- **CO 3:** Develop a detailed project plan by defining objectives, setting timelines, and identifying key deliverables to guide the implementation process.
- **CO 4:** Design and implement a prototype or initial model based on the proposed solution framework using appropriate AI tools and technologies.
- **CO 5 :** Demonstrate teamwork, communication, and project management skills by preparing and presenting a well-structured project proposal and initial implementation results

CO-PO-PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO 2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO 3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO 5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation: “-”

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

The TicketVerse – Event Ticket Booking System is a web-based application developed to provide a centralized and efficient platform for booking tickets for various events such as concerts, sports matches, theatre performances, and festivals. In today's fast-paced digital world, users expect quick and convenient access to services, including event booking. Traditional ticket booking methods, such as manual counters or unorganized online platforms, are often time-consuming, inefficient, and lack transparency. The proposed system eliminates these limitations by offering an interactive and user-friendly interface that allows users to browse available events, filter them based on categories, view detailed information, and book tickets seamlessly. The system includes features such as real-time event display, cart management, secure checkout, and user profile handling. The frontend of the system is developed using HTML, CSS, and JavaScript to provide a responsive and visually appealing interface, while the backend is implemented using Node.js to handle application logic and server-side operations. SQLite database is used to store event details, user information, and booking records securely. The system ensures data accuracy, reduces manual workload, and enhances user experience by providing a smooth and efficient ticket booking process. It is scalable and can be extended to support advanced features such as online payment integration, seat selection, and personalized recommendations.

LIST OF ABBREVIATIONS

SAMS – Student Attendance Management System

DB – Database

SQL – Structured Query Language

API – Application Programming Interface

UI – User Interface

HTML – HyperText Markup Language

JS – JavaScript

HTTP – HyperText Transfer Protocol

SMTP – Simple Mail Transfer Protocol

CRUD – Create, Read, Update, Delete

CHAPTER 1

INTRODUCTION

1.1 General

Event management and ticket booking play a crucial role in the entertainment and sports industries. With the increasing number of events being conducted worldwide, there is a growing demand for efficient and reliable ticket booking systems.

Traditional methods of ticket booking involve physical counters or third-party agents, which may lead to long waiting times, limited accessibility, and chances of human errors. With advancements in web technologies, online ticket booking systems have become a preferred solution. The TicketVerse system is designed to provide users with a simple, fast, and secure platform to explore events and book tickets from anywhere at any time.

1.2 Need for the System

The need for this system arises due to several limitations in existing systems:

- Manual booking systems are slow and inefficient
- Lack of centralized event information
- Difficulty in tracking bookings
- No proper user management
- Limited access to event details

The proposed system addresses these issues by:

- Providing an online platform accessible 24/7
- Allowing users to view and book events instantly
- Maintaining records in a structured database
- Reducing human errors and duplication
- Improving transparency and reliability.

1.3 Scope of the System

The scope of the TicketVerse system includes:

- Displaying various categories of events (Music, Sports, Theatre, Festivals)
- Allowing users to search and filter events
- Enabling ticket booking through a cart system
- Providing a secure checkout process
- Maintaining user profiles and booking history

The system can be used in:

- Colleges and universities for event management
- Entertainment industries
- Sports organizations
- Cultural event platforms

1.4 Organization of the Report

The project report is organized into several chapters to provide a clear and systematic understanding of the TicketVerse Event Ticket Booking System.

Chapter 1: Introduction
This chapter provides an overview of the system, including the general idea, need for the system, and its scope. It explains the motivation behind developing an online ticket booking platform and highlights the problems in traditional systems.

Chapter 2: System Design
This chapter explains the overall design of the system. It includes system architecture, data flow, and design considerations. It also describes how different components such as frontend, backend, and database interact with each other.

Chapter 3: System Implementation

This chapter focuses on the development process of the system. It explains the technologies used and provides detailed descriptions of each module such as Events, Cart, Checkout, Profile, and Ticket Management.

Chapter 4: Results and Discussion

This chapter presents the output of the system and evaluates its performance. It discusses how efficiently the system works, the advantages it provides, and any limitations observed during testing.

Chapter 5: Conclusion and Future Work

This chapter summarizes the overall work done in the project. It also suggests possible future enhancements such as adding payment gateways, mobile support, and advanced features.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Literature survey is an important phase in software development that involves studying existing systems, technologies, and research related to the proposed project. It helps in understanding current solutions, identifying their limitations, and finding ways to improve them.

In the context of event ticket booking systems, several online platforms and applications have been developed to simplify the process of booking tickets. These systems aim to provide users with easy access to event information, reduce manual effort, and improve overall efficiency.

This chapter discusses existing ticket booking systems, related technologies, and their advantages and limitations. Based on this study, the proposed TicketVerse system is designed to overcome the identified issues and provide an improved user experience.

2.2 Existing Ticket Booking Systems

Existing ticket booking systems include both traditional and modern digital platforms. Traditional systems involve booking tickets through physical counters or agents, which are simple but time-consuming and often lead to long queues and manual errors. With the advancement of technology, online ticket booking platforms and mobile applications have become more popular, allowing users to browse events, check availability, and book tickets from anywhere. These systems provide features such as real-time updates, digital payments, and electronic ticket confirmation. However, despite these advantages, many existing systems suffer from issues like complex user interfaces, dependency on internet connectivity, limited personalization, and security concerns during online transactions.

2.3 Related Work

Several research works and applications have focused on improving ticket booking systems using different technologies.

- Web-based booking systems have been widely used to provide centralized platforms for users.
- Cloud-based systems help in storing and managing large amounts of data efficiently.
- Some systems integrate AI to recommend events based on user preferences.
- Payment gateway integration is used to enable secure transactions.

Studies show that digital booking systems significantly reduce manual effort and improve user satisfaction. However, many systems still face challenges such as poor user experience, lack of personalization, and limited scalability

2.4 Summary

From the literature survey, it is evident that modern ticket booking systems have significantly improved the efficiency and convenience of booking processes compared to traditional methods. Online platforms and mobile applications enable users to access event information, book tickets instantly, and receive confirmations digitally, thereby reducing time and effort. However, existing systems still face certain challenges such as complex user interfaces, security concerns, dependence on internet connectivity, and limited personalization features. The proposed TicketVerse system aims to overcome these limitations by providing a simple, user-friendly interface, efficient event browsing, and a secure and reliable booking process, thereby enhancing overall user experience and system effectiveness.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In today’s digital era, booking tickets for events is still not fully efficient in many cases. Traditional ticket booking methods, such as physical counters and manual registrations, are time-consuming, prone to errors, and inconvenient for users. Even though online ticket booking systems are available, many of them suffer from issues such as complex user interfaces, lack of proper organization of events, and difficulty in managing bookings.

Users often face problems like difficulty in searching for events, lack of clear information, and complicated checkout processes. In addition, some systems do not provide proper ticket tracking or user profile management, which leads to poor user experience. Security concerns during transactions and inefficient data handling also affect the reliability of existing systems.

Therefore, there is a need for a simple, efficient, and user-friendly ticket booking system that allows users to easily browse events, book tickets, and manage their bookings without any complexity. The proposed TicketVerse system aims to address these issues by providing a centralized and well-structured platform for event ticket booking.

3.2 Objectives of the System

The main objective of the TicketVerse system is to develop a web-based application that simplifies and enhances the process of event ticket booking. The system aims to provide a smooth and efficient experience for users while ensuring accuracy and reliability.

The specific objectives of the system include:

- To provide a centralized platform for displaying various events in different categories
- To enable users to search and filter events easily
- To allow users to add tickets to a cart and manage their selections

- To implement a secure and user-friendly checkout process
- To store user details and booking information in a structured database
- To provide users with access to their booked tickets and history
- To reduce manual effort and eliminate errors in ticket booking
- To ensure fast response time and smooth system performance
- To design a responsive and interactive user interface
- To make the system scalable for future enhancements

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CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is an essential phase in software development where the overall structure, components, and workflow of the system are defined. It acts as a blueprint for the implementation phase and ensures that the system functions efficiently and meets user requirements.

The TicketVerse Event Ticket Booking System is designed as a web-based application that allows users to browse events, select tickets, and complete bookings in a simple and efficient manner. The system focuses on providing a user-friendly interface, fast performance, and secure data handling. A modular design approach is followed to ensure flexibility, maintainability, and scalability.

4.2 System Architecture

The TicketVerse system follows a three-tier architecture, which divides the system into three main layers:

1. Presentation Layer

This layer represents the user interface of the system.

- Developed using HTML, CSS, and JavaScript
- Displays event listings, forms, and user dashboards
- Provides interactive features such as search, filtering, and navigation
- Ensures responsive design for better user experience

2. Application Layer

This layer handles the business logic and processing.

- Implemented using Node.js
- Processes user requests such as ticket booking and cart management
- Validates user inputs
- Communicates with the database layer

3. Database Layer

This layer manages data storage and retrieval.

- Uses SQLite database
- Stores event details, user information, and booking records
- Ensures data consistency and security
- Supports fast data access and updates

4.3 System Requirements

System requirements define the resources needed for the system to function properly.

4.3.1 Hardware Requirements

The system can run on standard computing devices without requiring high-end hardware.

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB HDD or SSD
- Input Devices: Keyboard and Mouse
- Internet Connection

4.3.2 Software Requirements

The system is developed using widely available software technologies.

- Operating System: Windows 10 or higher
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: Node.js
- Database: SQLite
- Development Environment: Visual Studio Code
- Web Browser: Google Chrome or any modern browser

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CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The TicketVerse Event Ticket Booking System is implemented using a web-based development approach that allows users to access the application through a browser. The development process follows a structured methodology that includes requirement analysis, system design, implementation, testing, and deployment.

The frontend of the system is developed using HTML, CSS, and JavaScript to create an interactive and responsive user interface. The backend is implemented using Node.js, which handles server-side logic, processes user requests, and manages communication with the database. SQLite is used as the database to store event details, user information, and booking records.

The system follows a modular approach where each functionality is implemented as a separate module. This improves maintainability, allows easy debugging, and supports future enhancements.

5.2 Module Descriptions

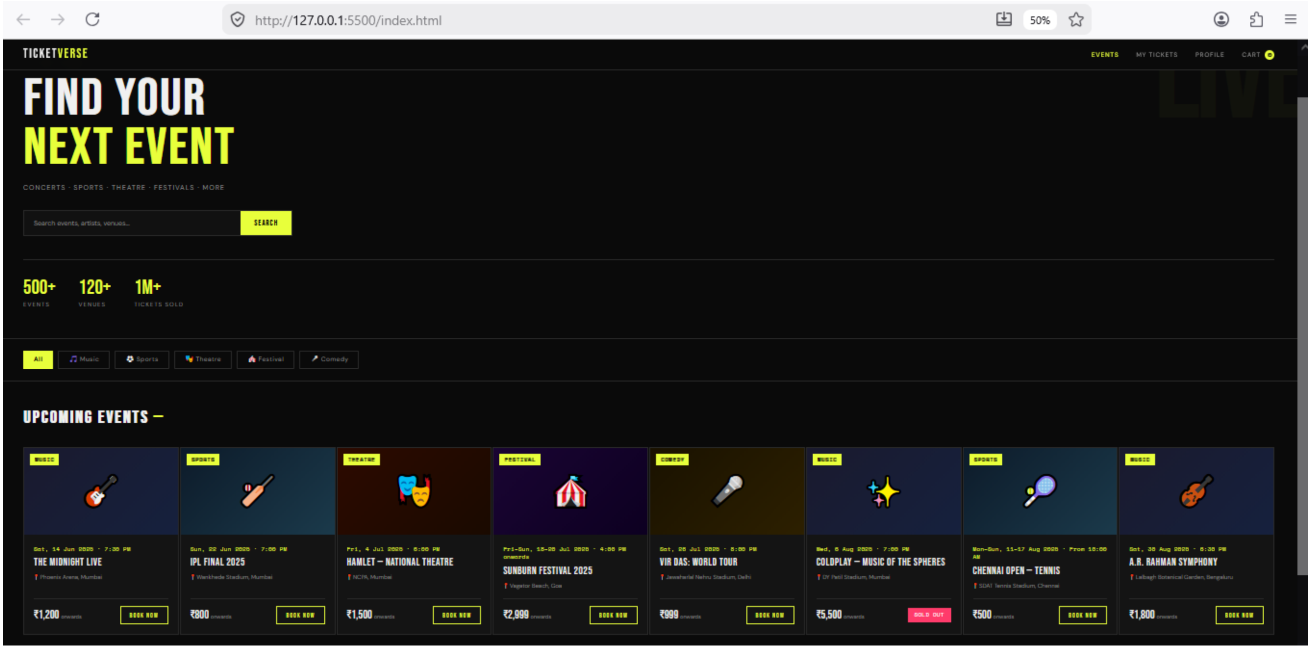
The TicketVerse system consists of several modules that work together to provide efficient ticket booking functionality.

5.2.1 Home / Events Module

The Home module acts as the entry point of the system. It displays all available events in a structured format.

- Shows different categories such as Music, Sports, Theatre, and Festivals
- Allows users to search and filter events
- Displays event details like date, time, and venue

This module improves user engagement by providing a clear and attractive interface.

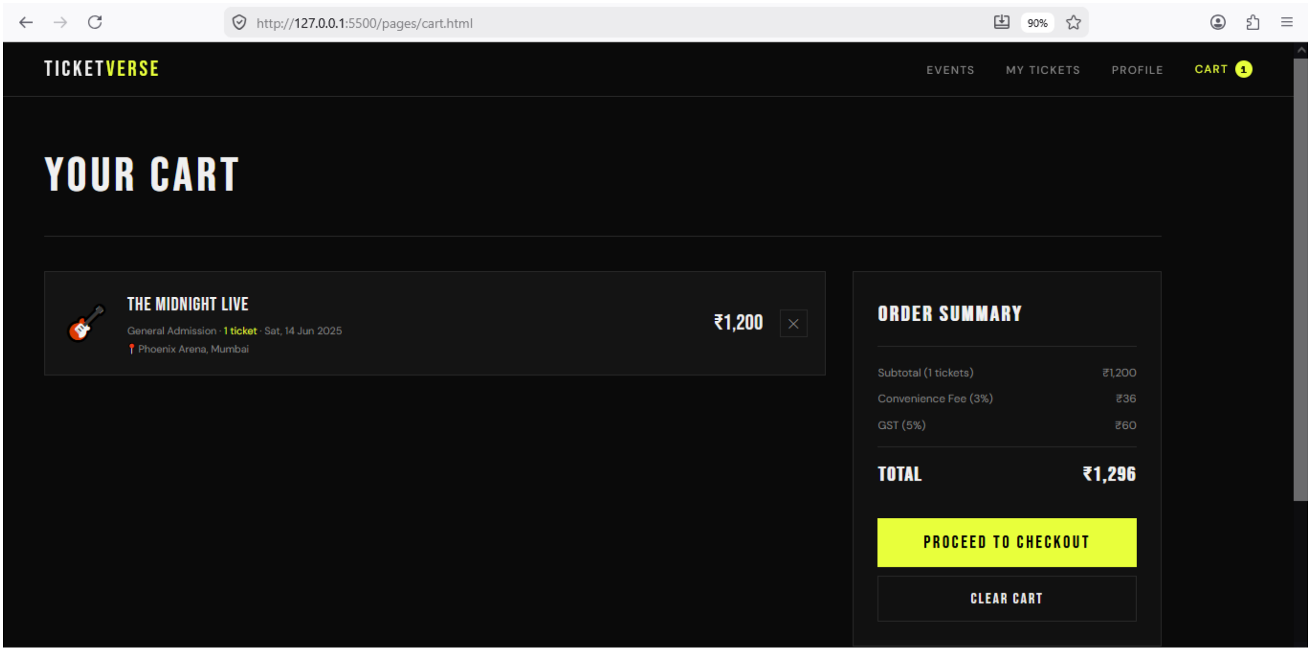


5.2.2 Cart Module

The Cart module is used to manage selected tickets before final booking.

- Allows users to add tickets to the cart
- Displays the number of selected items
- Enables users to remove or update tickets

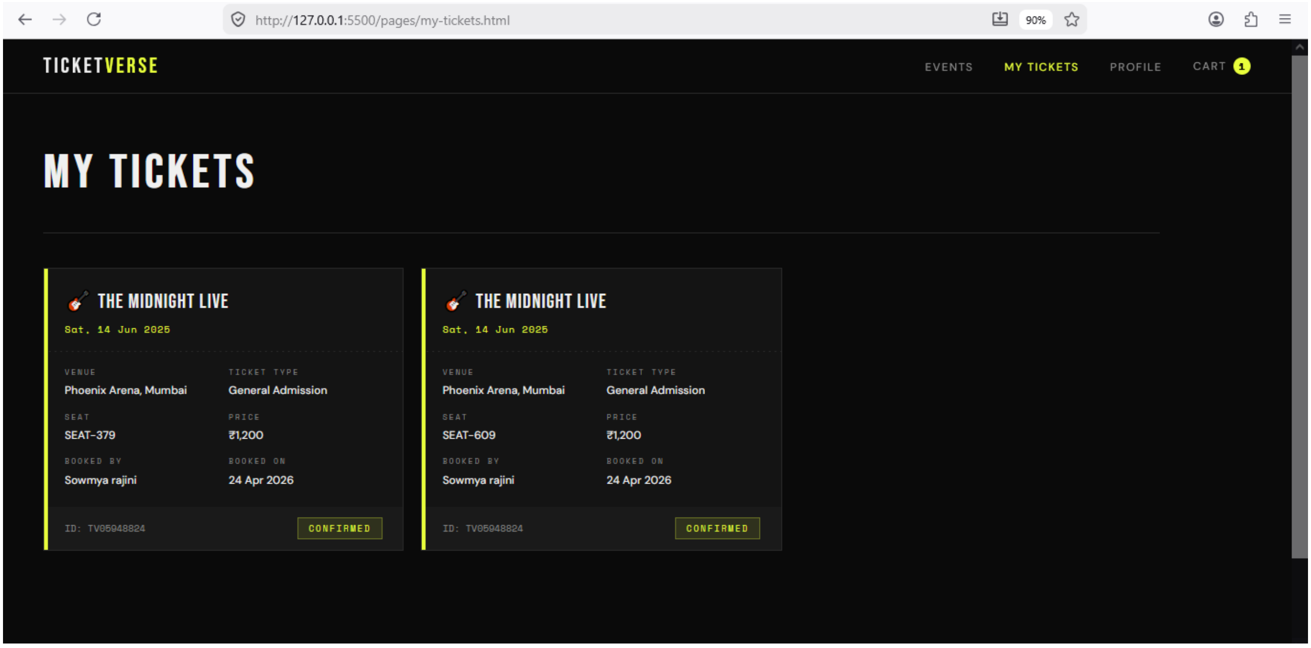
This module ensures flexibility in the booking process and prevents errors before checkout.



5.2.3 My Tickets Module

This module allows users to view their booked tickets.

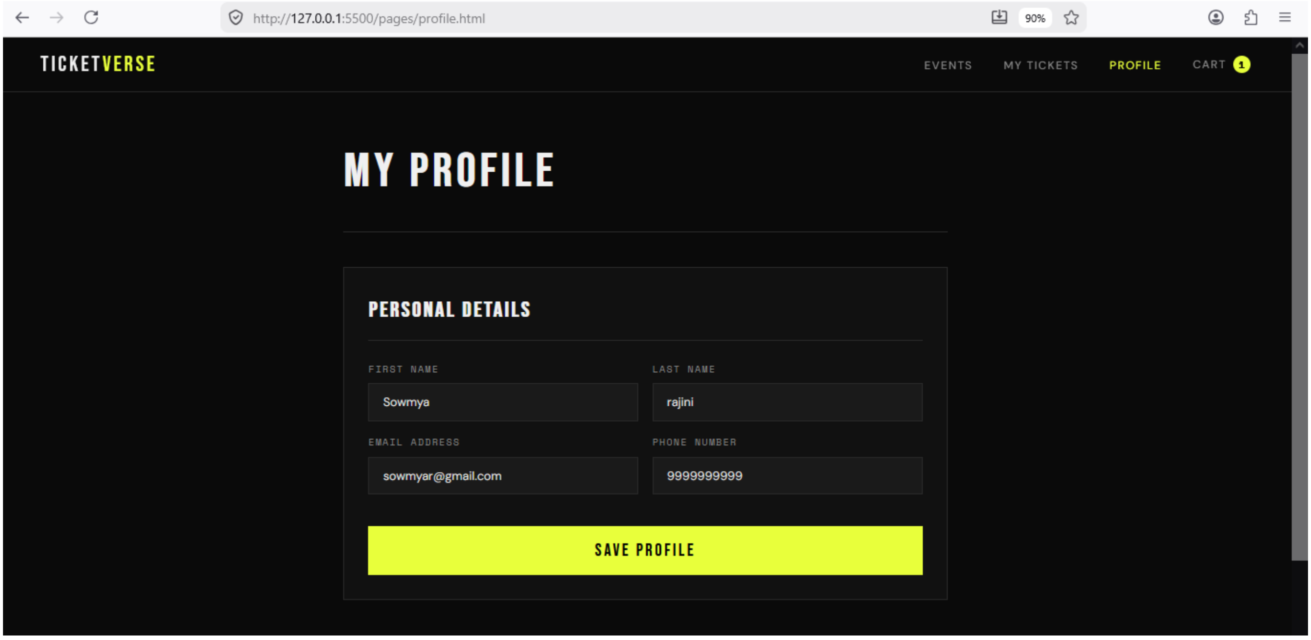
- Displays all previous bookings
- Shows event details and booking status
- Helps users track their tickets easily



5.2.4 Profile Module

The Profile module manages user information.

- Stores personal details of the user
- Allows editing and updating of information
- Helps in faster checkout during future bookings



5.2.5 Admin Module

This module provides administrative control over the system.

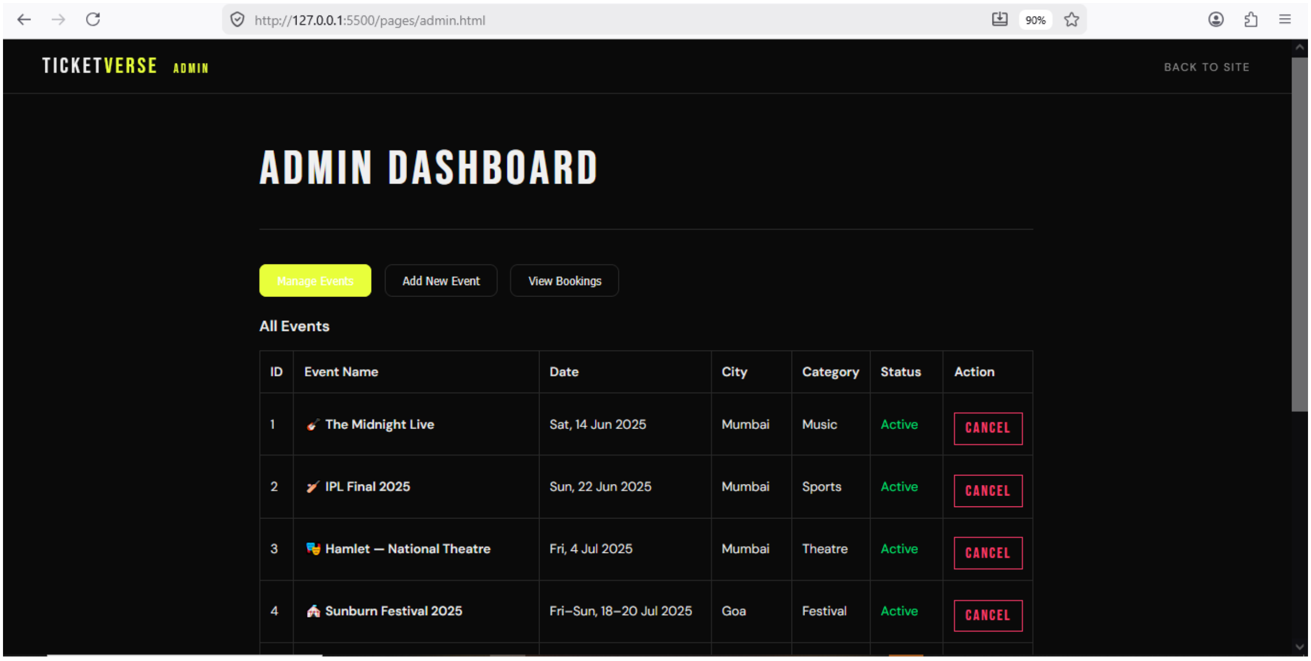
• Manages all event-related operations

• Allows adding, updating, and deleting events

• Enables viewing of all user bookings

• Controls event status (active/inactive)

• Ensures proper organization of system data



5.3 Implementation Features

The system includes several important features:

- User-friendly interface for easy navigation
- Real-time event browsing and filtering
- Cart system for managing selected tickets
- Secure checkout process
- Ticket tracking system
- Efficient database management

5.4 Advantages of Implementation

- Reduces manual effort in booking
- Saves time for users
- Improves accuracy and reliability
- Provides better user experience
- Easy to maintain and update

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The TicketVerse Event Ticket Booking System was successfully developed and tested as a web-based application. The system provides a smooth and interactive platform where users can browse events, select tickets, and complete bookings efficiently.

During execution, the system performed all core functionalities correctly. Users were able to view different categories of events, access detailed information about each event, and add tickets to the cart without any issues. The checkout process was completed successfully, and booking details were stored accurately in the database.

The system also allowed users to view their booked tickets through the “My Tickets” module, which displayed all relevant details such as event name, date, and booking status. The profile module enabled users to manage their personal information effectively.

Overall, the system achieved its objective of simplifying the ticket booking process and providing a user-friendly experience.

6.2 Performance Evaluation

The performance of the TicketVerse system was evaluated based on usability, efficiency, and response time.

- **Usability:** The system provides a simple and intuitive interface that allows users to navigate easily between different modules such as events, cart, and checkout. Even first-time users can operate the system without difficulty.
- **Efficiency:** The system processes user requests quickly and performs operations such as adding tickets to cart, retrieving event data, and storing bookings without delay.
- **Response Time:** The use of Node.js and SQLite ensures fast data processing and quick response time during user interactions.
- **Scalability:** The modular design allows the system to be extended with additional features in the future without affecting existing functionality.

The system performs well under normal usage conditions and provides reliable results.

6.3 System Testing

The system was tested to ensure that all functionalities work correctly and meet the expected requirements.

- **Test Case 1: Event Browsing**
  Result: PASS – Events are displayed correctly with proper details
- **Test Case 2: Add to Cart**
  Result: PASS – Tickets are added and updated successfully
- **Test Case 3: Checkout Process**
  Result: PASS – User details are captured and booking is completed
- **Test Case 4: Ticket Viewing**
  Result: PASS – Booked tickets are displayed correctly
- **Test Case 5: Profile Management**
  Result: PASS – User details can be updated successfully

The testing results confirm that the system is functioning correctly and efficiently.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The TicketVerse Event Ticket Booking System was successfully designed and developed as a web-based application to simplify the process of booking tickets for various events. The system effectively replaces traditional booking methods with a digital platform that is more efficient, reliable, and user-friendly.

The application allows users to browse different categories of events, view detailed information, add tickets to a cart, and complete the booking process through a structured and interactive interface. All booking data is stored in a centralized database, ensuring accuracy, easy retrieval, and secure management of information.

The system reduces manual effort, minimizes errors, and improves the overall user experience by providing quick access to event details and a smooth booking process. The modular design of the system also ensures maintainability and flexibility for future improvements.

Overall, the project meets its objectives by delivering a functional and efficient ticket booking system that can be used in real-world applications.

7.2 Future Enhancements

Although the current system provides essential features for ticket booking, several enhancements can be made to improve its functionality and performance in the future:

- Integration of secure online payment gateways for real-time transactions
- Implementation of seat selection and seat availability features
- Development of a mobile application for better accessibility
- Addition of AI-based event recommendations based on user preferences
- Integration of email and SMS notifications for booking confirmation
- Implementation of user authentication with advanced security features

- Support for multi-language interface to reach a wider audience
- Integration with external event management systems

These enhancements will make the system more advanced, scalable, and suitable for large-scale deployment.

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APPENDIX
PSEUDOCODE

System Initialization Logic

START

Connect to database

IF connection successful

→ Start server (port 3000)

→ Load homepage (index.html)

ELSE

→ Show error message

END IF

END

User Login Logic

FUNCTION Login(email, password)

Check user table for matching details

IF valid

→ Create session

→ Redirect to homepage

ELSE

→ Show “Invalid Login”

END IF

END FUNCTION

Add to Cart Logic

FUNCTION AddToCart(event_id)

Check if event exists

IF exists

→ Add event to cart

→ Update cart count

ELSE

→ Show error

END IF

END FUNCTION

Checkout Logic

FUNCTION Checkout(user_details, cart_items)

Calculate total amount

Store booking details in database

IF successful

→ Clear cart

→ Show confirmation

ELSE

→ Show error message

END FUNCTION

Admin Event Management Logic

FUNCTION AddEvent(event_details)

Insert event into database

Return success message

END FUNCTION

FUNCTION DeleteEvent(event_id)

Remove event from database

END FUNCTION

View Tickets Logic

FUNCTION ViewTickets(user_id)

Fetch bookings from database

Display event details

END FUNCTION

SYSTEM TESTING

System Test Verification

Test Case 1: Event Display

Result: PASS – Events displayed correctly

Test Case 2: Add to Cart

Result: PASS – Items added successfully

Test Case 3: Checkout Process

Result: PASS – Booking completed

Test Case 4: Ticket Viewing

Result: PASS – Tickets displayed

Test Case 5: Admin Event Management

Result: PASS – Events added/removed successfully

Final System Status

The TicketVerse system has been successfully developed and tested. It provides efficient event browsing, ticket booking, and admin management. The system ensures smooth performance, accurate data handling, and user-friendly interaction.